

1. Administrative & Study Identification

Full Study Title: A Phase 3 Randomized, Open-label, Multicenter Study to Compare the Efficacy and Safety of Sacituzumab Tirumotecan (Sac-TMT, MK-2870) in Combination With Pembrolizumab Versus Pembrolizumab Alone as First-line Maintenance Treatment in Participants With Mismatch Repair Proficient Endometrial Cancer (TroFuse-033/GOG-3119/ENGOT-en29) **Short Title/Acronym:** TroFuse-033 / MK-2870-033 **Protocol Number:** MK-2870-033 **NCT Number:** NCT06952504 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Sacituzumab Tirumotecan (sac-TMT, MK-2870) and Pembrolizumab (MK-3475) **Phase of Development:** Phase 3 **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** Sponsor Medical Monitor (Merck)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if participants with mismatch repair proficient (pMMR) endometrial cancer who receive sacituzumab tirumotecan (sac-TMT) in combination with pembrolizumab live longer (Overall Survival, OS) and without the cancer getting worse (Progression-Free Survival, PFS) compared to those who receive pembrolizumab alone during the maintenance phase. **Secondary Objectives:** To evaluate Progression-Free Survival 2 (PFS2), Objective Response Rate (ORR), Duration of Response (DOR), Health-Related Quality of Life (HRQoL) using the EORTC QLQ-C30 scale, and overall safety and tolerability (including number of participants experiencing adverse events and discontinuations due to adverse events).

2.2 Background & Rationale Researchers are investigating new ways to treat people with proficient mismatch repair (pMMR) endometrial cancer (EC) that is advanced or recurrent. Sac-TMT is a TROP2-directed antibody-drug conjugate (ADC) that attaches to specific targets on cancer cells to deliver treatment and destroy them. Integrating immunotherapy (pembrolizumab) with this targeted ADC in the maintenance setting aims to improve upon standard-of-care clinical outcomes for patients whose cancer does not progress after initial induction chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab monotherapy) **Blinding:** Open-label **Randomization:** Randomized (1:1 ratio into maintenance arms) **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target \$\$\$: 1,123 participants (Estimated) **Number of Sites:** Approximately 239 locations globally **Estimated Study Start:** 22-May-2025 **Estimated Primary Completion:** 24-May-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has a histologically confirmed diagnosis of primary advanced or recurrent endometrial carcinoma that has been confirmed as proficient mismatch repair (pMMR). 2. Has radiographically evaluable disease, with measurable Stage III or either measurable or non-measurable Stage IV or recurrent disease per RECIST 1.1, as assessed by the investigator. 3. Has received no prior systemic therapy for endometrial carcinoma except pre-specified conditions (e.g., 1 prior line of systemic platinum-based adjuvant/neoadjuvant chemotherapy with curative-intent).

4.2 Exclusion Criteria 1. Has carcinosarcoma, neuroendocrine tumors, or endometrial sarcoma (including stromal sarcoma, leiomyosarcoma, adenosarcoma). 2. Has endometrial carcinoma of any histology that is mismatch repair deficient (dMMR). 3. Is a candidate for debulking surgery resulting in complete removal of all tumors, or curative-intent radiotherapy at the time of enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Induction Phase:** All participants receive pembrolizumab 200 mg, carboplatin (Trade name: Paraplatin) AUC 5, and paclitaxel 175 mg/m² or docetaxel 75 mg/m² on Day 1 of each 3-week cycle for 6 cycles. * **Maintenance Phase:** Non-progressing participants are randomized to either Arm A (Pembrolizumab + Sac-TMT) or Arm B (Pembrolizumab alone) on Day 1 of each 3-week cycle. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** 3-week cycles; timeframes extend up to approximately 44 months for PFS evaluation and 54 months for OS evaluation.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications to manage chemotherapy and ADC-related side effects. **Prohibited:** Prior therapy with anti-PD-1/PD-L1/PD-L2 agents, TROP2-targeted ADCs, or topoisomerase I inhibitor-containing ADCs. Concurrent investigational treatments are strictly forbidden.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent,

Medical History, Tumor biopsy/pMMR confirmation, Baseline RECIST 1.1 Imaging | | **Baseline** | Day 1 | Initiation of Induction Phase (First Dose of Pembrolizumab + Chemo) | | **Interim** | Every 3 Weeks | IV Infusions, Safety Labs, AE Assessment; Radiographic tumor imaging scheduled at specified intervals | | **End of Study** | End of Treatment | Final Radiographic Imaging, Survival Follow-Up, Safety Exit Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: * **Progression-Free Survival (PFS):** Assessed in the Maintenance Treatment phase, defined as the time from randomization to the first documented progressive disease (PD) or death due to any cause, whichever occurs first. * **Overall Survival (OS):** Assessed in the Maintenance Treatment phase, defined as time from randomization to death due to any cause. **Measurement Method:** Blinded Independent Central Review (BICR) using RECIST 1.1 criteria.

7.2 Secondary & Exploratory Endpoints * **Progression-Free Survival 2 (PFS2):** Maintenance Treatment PFS2 as assessed by the Investigator. * **Safety Outcomes:** Maintenance Treatment: Number of Participants Who Experience One or More Adverse Events (AEs), and Number of Participants Who Discontinue Study Intervention Due to an AE. * **Objective Response Rate (ORR)** and **Duration of Response (DOR).** * **Change from baseline in Health-Related Quality of Life (HRQoL)** utilizing the EORTC QLQ-C30 scale.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of AEs from the first dose through standard oncology follow-up. Active monitoring for immune-related adverse events (irAEs) and specific ADC-related toxicities (e.g., severe dry eye syndrome, severe corneal disease, and inflammatory bowel disease). Number of participants with AEs and AE-driven study discontinuations are tracked as key secondary outcomes. **Serious Adverse Events (SAEs):** Protocol-mandated 24-hour reporting timeframe to the sponsor. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) is established to actively evaluate trial safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy endpoints (PFS/OS). Safety Analysis Set for AE/SAE frequency reporting. **Sample Size Justification:** $N = 1,123$ statistically powered ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to determine superiority in PFS and OS between the combination therapy and monotherapy arms. **Handling of Missing Data:** Standard

Kaplan-Meier censoring rules for time-to-event endpoints (PFS and OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits managed by the sponsor to assure protocol adherence. **Audit Trail:** Mandatory eCRF tracking, comprehensive centralized data clarification processes, and BICR validation for all radiographic images.

11. Study Identifiers & Governance

Primary ID: MK-2870-033 **Posting ID:** 784571079295545450 **Registry Number:** NCT06952504 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (Merck) **Study Title:** TroFuse-033 **Official Regulatory Title:** A Phase 3 Randomized, Open-label, Multicenter Study to Compare the Efficacy and Safety of Sacituzumab Tirumotecan (Sac-TMT, MK-2870) in Combination With Pembrolizumab Versus Pembrolizumab Alone as First-line Maintenance Treatment in Participants With Mismatch Repair Proficient Endometrial Cancer

12. Clinical Framework (Oncology Specific)

Primary Condition: Endometrial Cancer (Mismatch Repair Proficient Endometrial Cancer) **Diagnosis Threshold:** Advanced or recurrent pMMR Endometrial Carcinoma **Medical Methodology:** Immune Checkpoint Inhibitor combined with a TROP2-directed Antibody-Drug Conjugate **Optimization Target:** Delay of disease progression (PFS) and extended Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard induction chemotherapy followed by active maintenance randomization. **Education Protocol (HCP):** Specialized site visits for ADC toxicity management, RECIST 1.1 documentation, and irAE management. **Education Protocol (Patient):** Strict onsite training for early recognition of ophthalmic, gastrointestinal, or pulmonary symptoms. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all imaging to ensure unbiased progression determinations.

14. Observation & Follow-Up Schedule

Total Duration: Estimated 7 years (completion targeted May 2032)

Onsite Visit Cadence: Day 1 of every 3-week cycle (Q3W) **Remote**

Monitoring Frequency: Inter-cycle as clinically indicated for safety evaluations. **Checklist Review Interval:** Radiographic tumor imaging intervals strictly bound to protocol timelines (typically every 9-12 weeks) to evaluate objective RECIST 1.1 responses.

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					